Modeling the Molecular Distance Geometry Problem Using Dihedral Angles

Michael Souza

Federal University of Ceará/Brazil

Department of Applied Mathematics

Carlile Lavor

Univerisity of Campinas

Department of Applied Mathematics

Definitions

Distance Geometry Problem

Distance Geometry (DG) *is the study of geometry based on distance.*

- *Menger, Dimension theorie. Teubner. Berlim, 1928.*
- *Blumenthal, Theory and applic. of distance geometry. Oxford, 1953.*
- *Yemini, The positioning problem, Proc. Dist. Sensor Networks, 1978.*

The ***Distance Geometry Problem*** (DGP) is to find points

$$x_1, x_2, \ldots, x_n \in V,$$

such that

$$l_{ij} \leq \|x_i - x_j\| \leq u_{ij}, \quad (i,j) \in E \subseteq \{1,2,\ldots,n\}^2.$$

Distance Geometry Problem

The ***Distance Geometry Problem*** (DGP) is to find points

$$x_1, x_2, \ldots, x_n \in V,$$

such that

$$l_{ij} \leq \|x_i - x_j\| \leq u_{ij}, \quad (i,j) \in E \subseteq \{1,2,\ldots,n\}^2.$$

Esparse

Input

Incomplete and imprecise list of distances

$l_{ij} < u_{ij}$

Ouput

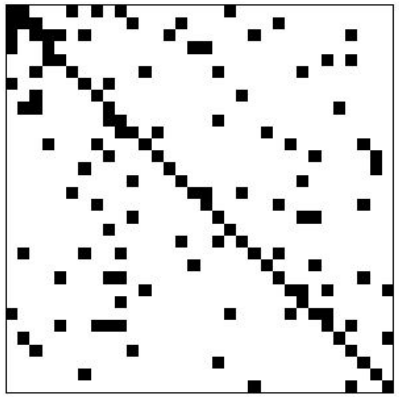
A set of points in some Euclidean space satisfying this constraints

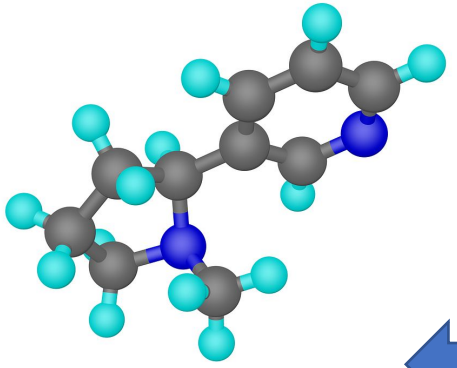
Complexity

Distance Geometry Problem

- *Saxe, Embeddability of weighted graphs in k-space is strongly NP-hard. Proc. 17th Allerton Conf. in Comm., Control, and Computing, 1979.*

Distance Geometry Problem





Partition Problem

$$S = \{1,2,5,6,7,8,9\}$$

$$S_1 = \{1,7,8\} \qquad S_2 = \{2,5,9\}$$

DGP is a NP-Hard problem

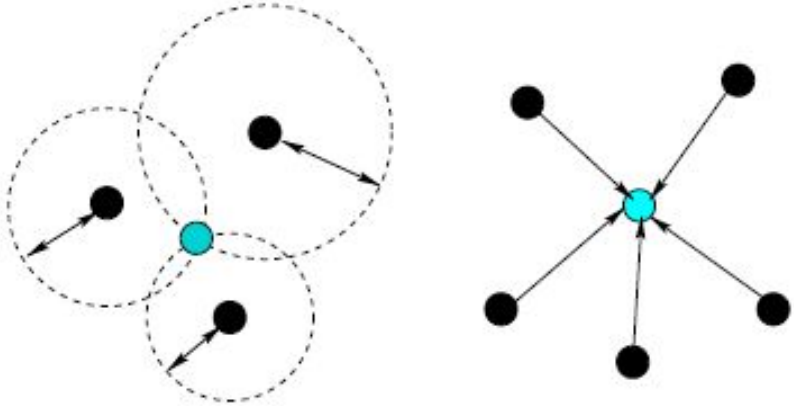
Applications

DGP - Applications

Wireless sensor networks *(without GPS)*

- Localization of network sensors

- *Savvides et al., Dynamic fine-grained loc. in ad-hoc nets of sensors, 2001.*
- *Eren et. al, Rigidity, computation, and randomization in net loc., 2004.*
- *Liu et al., Survey of wireless indoor positioning techniques and sys., 2007.*



Images from Savvides2001

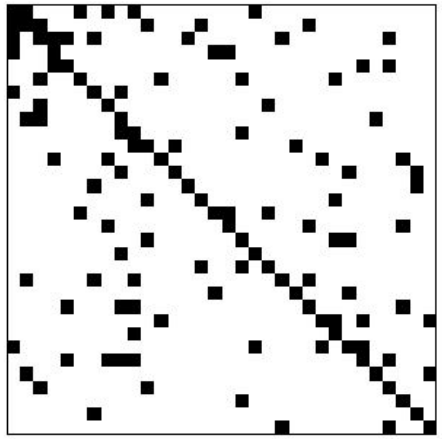
DGP - Applications

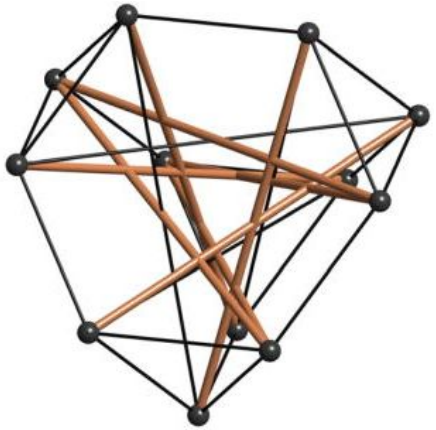
Statics

- Equilibrium under the action of external forces.

- *Alfakih, Graph rigidity via Euclidean distance matrices, 2000.*
- *Alfakih, On dimensional rigidity of bar-and-joint frameworks, 2007.*

***Rigidity** and **uniqueness** of solution*





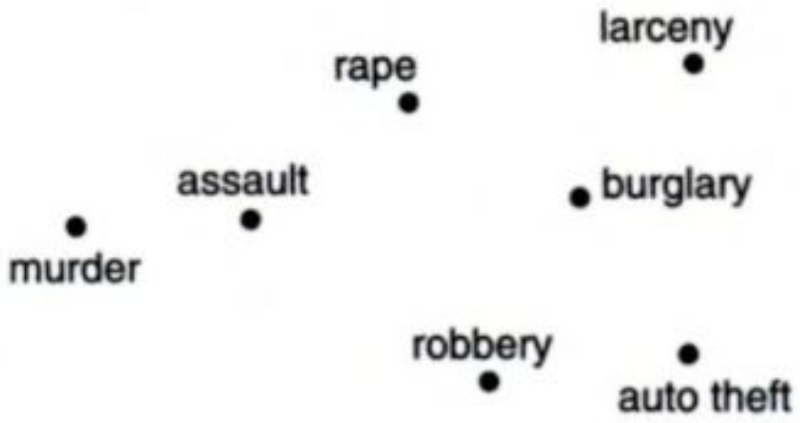
DGP - Applications

Dimensionality Reduction

- Multidimensional scaling (MDS)

- *Borg et al., Modern multidimensional scaling: Theory and appl., 2010.*
- *Everitt et al., The analysis of proximity data, 1997.*
- *Dzemyda et al., Multidimensional data visualization, 2012.*

Crime	No.	1	2	3	4	5	6	7
Murder	1	1.00	0.52	0.34	0.81	0.28	0.06	0.11
Rape	2	0.52	1.00	0.55	0.70	0.68	0.60	0.44
Robbery	3	0.34	0.55	1.00	0.56	0.62	0.44	0.62
Assault	4	0.81	0.70	0.56	1.00	0.52	0.32	0.33
Burglary	5	0.28	0.68	0.62	0.52	1.00	0.80	0.70
Larceny	6	0.06	0.60	0.44	0.32	0.80	1.00	0.55
Auto theft	7	0.11	0.44	0.62	0.33	0.70	0.55	1.00



Images and data from Borg2010.

DGP - Applications

Nanostructures

- Unlabeled distances

- *Juhás et.al, Ab initio determin. of solid-state nanostruct., Nature, 2006.*
- *Billinge et. al, Assigned and unassigned distance geometry, 4OR, 2016.*

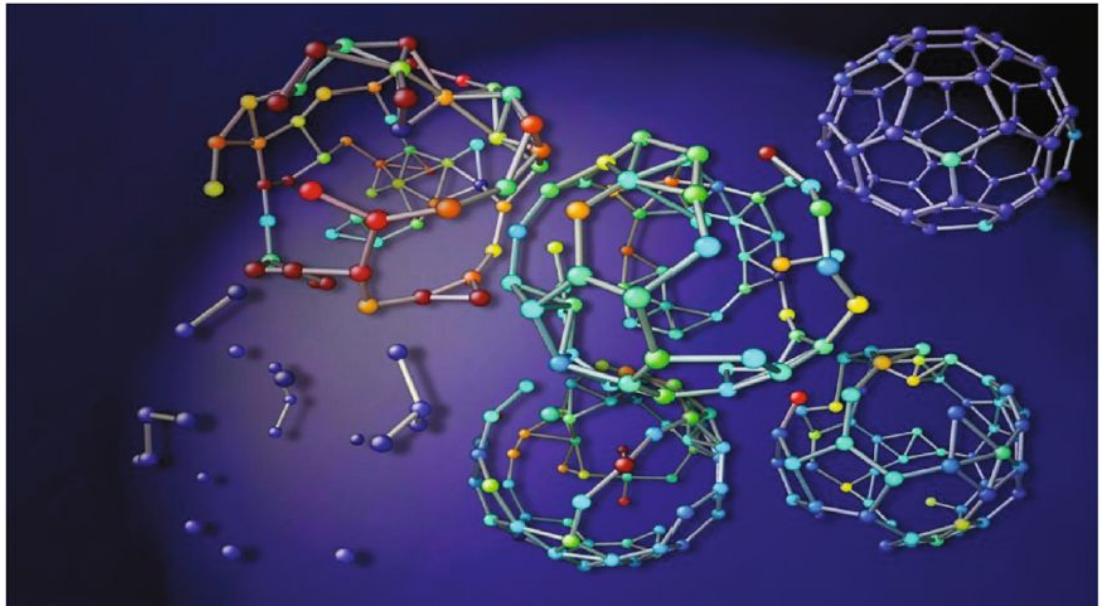


Image from Billinge2016.

DGP - Applications

NMR

- Protein structure determination

Method	***#Proteins***
X-Rray	105,656
NMR	10,257
Electron Microscopy	993
Hybrid	97
Other	181
Total	***117,184***

(From: PDB - Accessed in 01/30/2017).

DGP - Applications

NMR

- Protein structure determination

NMR doesn't give positions, but only info about the distance between some pairs of atoms

$$l_{ij} \leq \|x_i - x_j\| \leq u_{ij},$$

$$(i,j) \in E \subseteq \{1,2,\dots,n\}^2$$

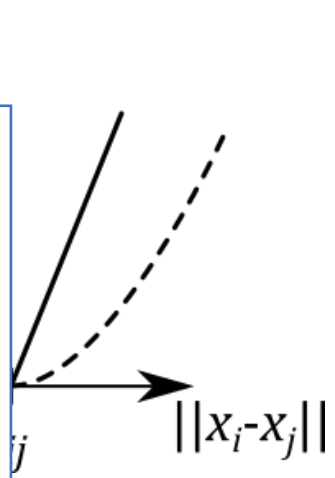
Algorithms and the underlying hypothesis

DGP - Algorithms

- **Global Optimization – General Methods (No additional hypothesis)**

$min_x\ f(x$

$p_{ij}(x) =$



Challenge:

- **The number of local minima grows exponentially with the problem size;**

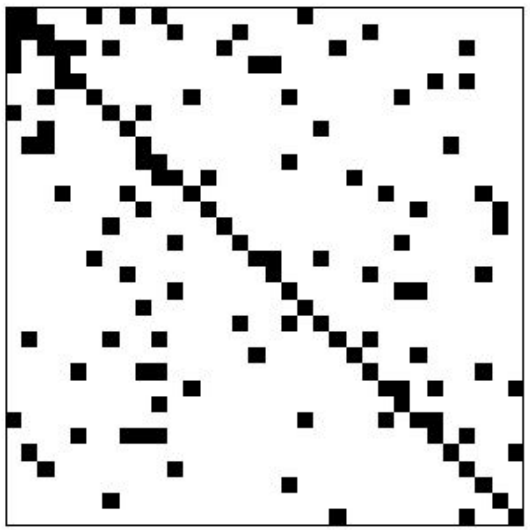
- *Moré a*
- *An and* *oach, 2003.*
- *Biswas and Ye, SDP for ad hoc wireless sensor network localization, 2004.*
- *Souza et. al. Solving the MDGP with inaccurate distance data. BMC, 2013.*

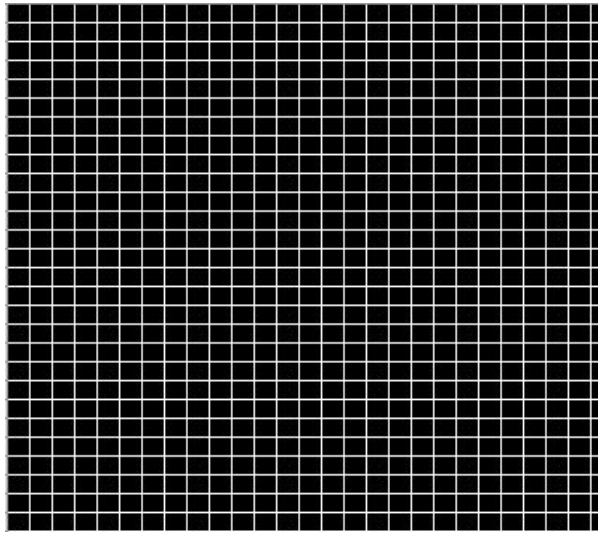
DGP - Algorithms

Geometric build-up – Additional hypothesis

$H_1: (i,j) \in E, \forall i,j;$

$H_2: l_{ij} = u_{ij}, \forall (i,j) \in E;$





$DGP\ can\ be\ solved\ by\ SVD\ in\ O(n^3)\ operations.$

Crippen and Havel, DG and Molecular Conformation, 1988.

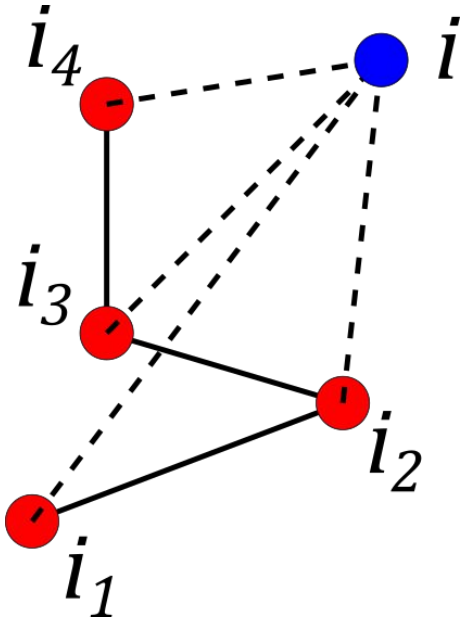
DGP - Algorithms

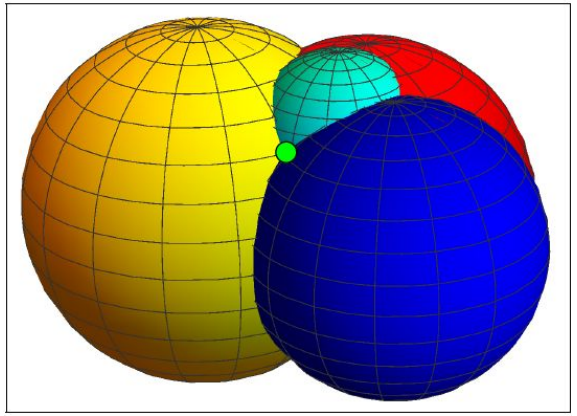
Geometric build-up – Additional hypothesis

H_1: $\forall i > 4$, *exist at least four atoms* $i_1, i_2, i_3, i_4 < i$, *such that*

$$(i, i_k) \in E, k = 1, 2, 3, 4;$$

H_2: $l_{ij} = u_{ij}, \forall (i, j) \in E;$





DGP can be solved in $O(n)$ operations.

Dong and Wu, A linear-time algorithm for solving the MDGP with exact inter-atomic distances, 2002.

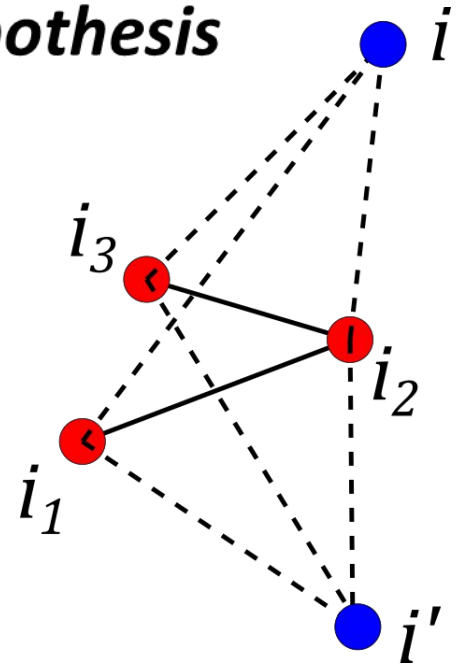
DGP - Algorithms

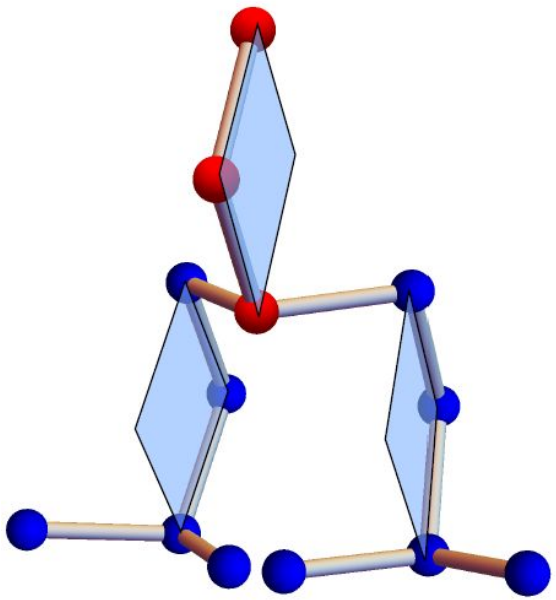
Geometric build-up – Additional hypothesis

H_1: $\forall i > 3$, *exist at least three atoms* $i_1, i_2, i_3 < i$, *such that*

$$(i, i_k) \in E, k = 1, 2, 3;$$

H_2: $l_{ij} = u_{ij}, \forall (i,j) \in E;$





Find all solutions of DGP.
In practice, $O(n)$ operations, but $O(n^2)$ in worst case.

Liberti et. al, A branch-and-prune algorithm for the MDGP. ITORS, 2008.

DGP - Algorithms

Geometric build-up – Additional hypothesis

H_1: $\forall i > 3$, *exist at least three atoms* $i_1, i_2, i_3 < i$, *such that*

$$(i, i_k) \in E, k = 1, 2, 3;$$

H_2: $l_{ij} = u_{ij}, \forall (i, j) \in E;$

Are these hypothesis still too strong?

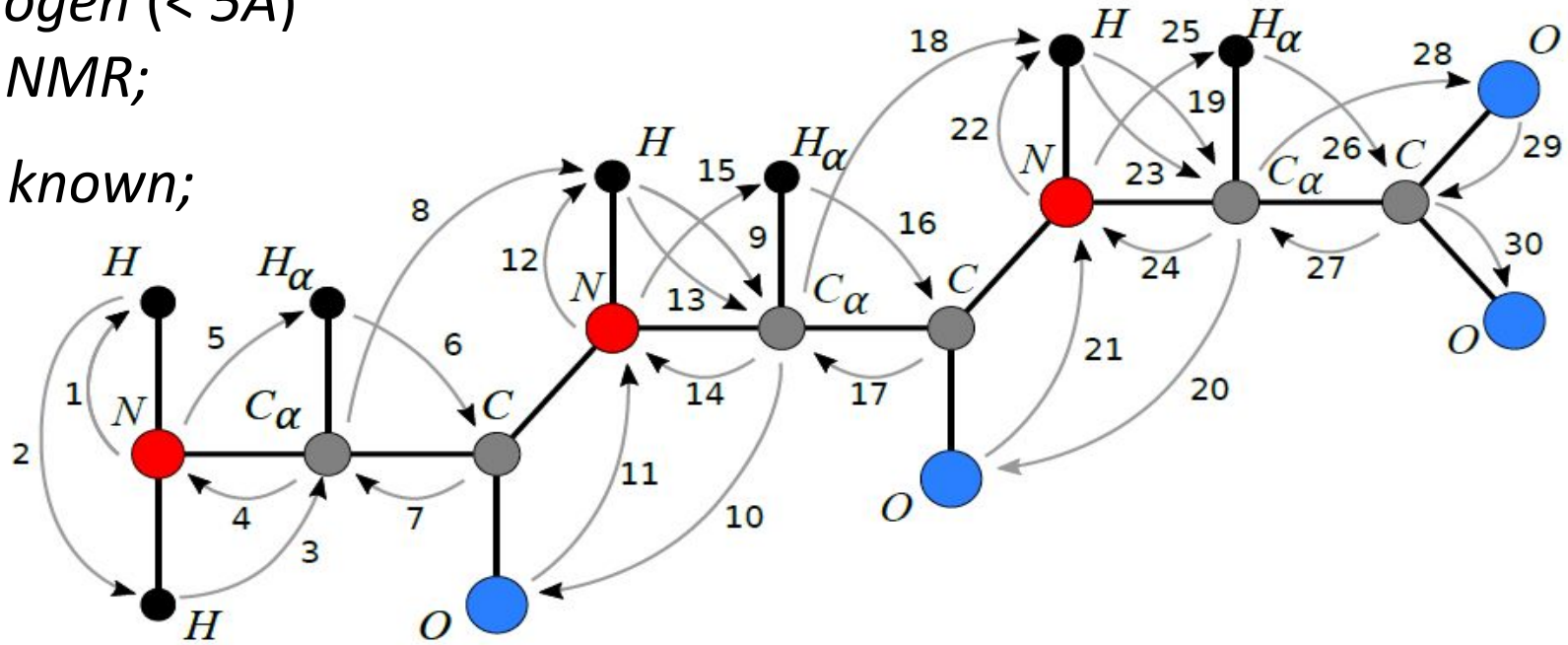
For protein backbones, what are the guaranteed valid hypothesis?

DGP - Algorithms

Geometric build-up – Additional hypothesis

Assumptions

- *Distances between close hydrogen (< 5A) atoms will be estimated by NMR;*
- *Length of covalent bonds are known;*
- *Planar angles are known;*



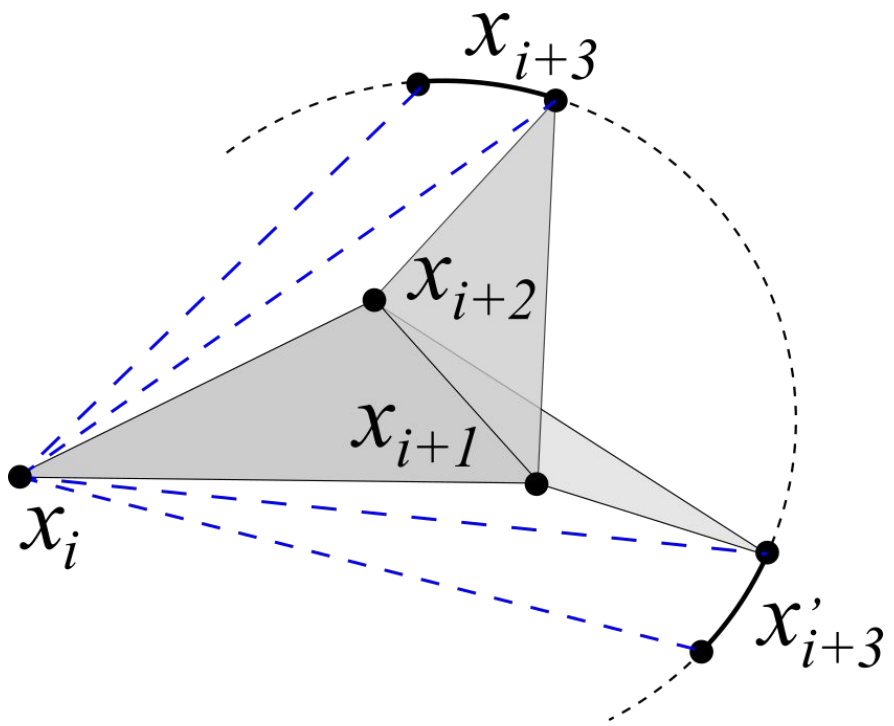
Lavor et. al, On the computation of protein backbones by using artificial backbones of hydrogens. Journal of Global Optimization, 2011.

DGP - Algorithms

Geometric build-up – Additional hypothesis

$H_1: (i,j) \in E, \forall |i-j| < 4;$

$H_2: l_{ij} = u_{ij}, \forall |i-j| < 3;$



Lavor et. al, On the computation of protein backbones by using artificial backbones of hydrogens. Journal of Global Optimization, 2011.

DGP - Algorithms

Geometric build-up – Additional hypothesis

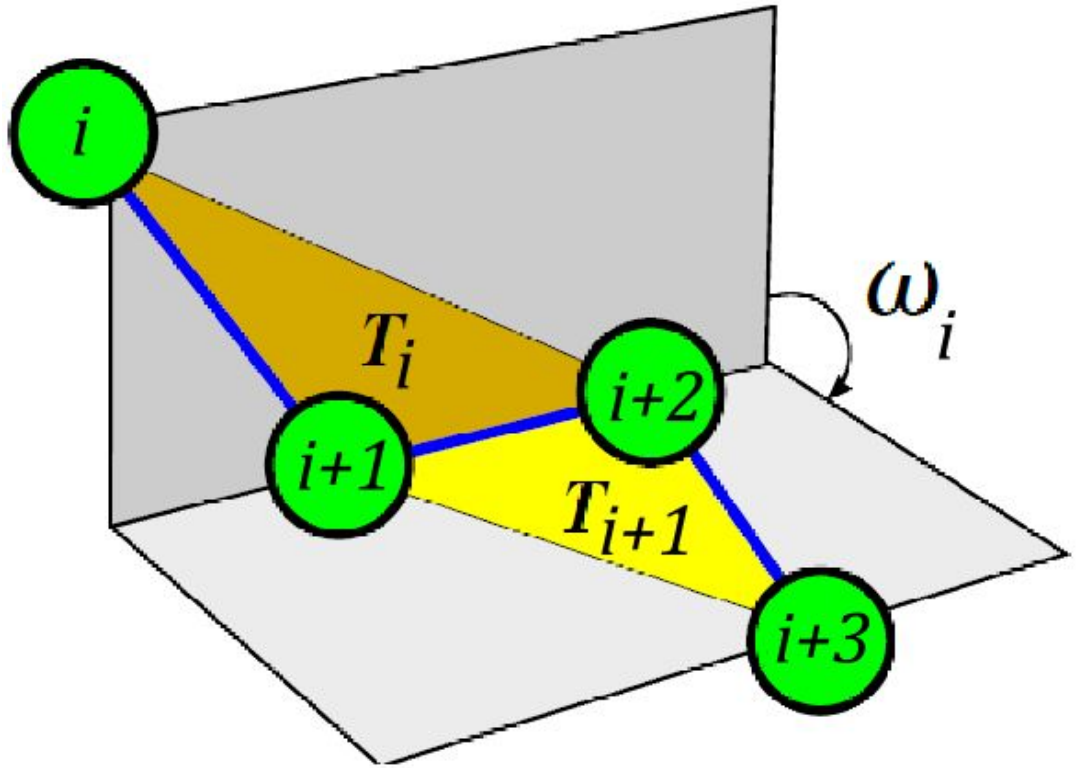
$H_1: (i,j) \in E, \forall |i-j| < 4;$

$H_2: l_{ij} = u_{ij}, \forall |i-j| < 3;$

The only degree of freedom is given by the dihedral (torsion) angles;

$$x_k = R(\omega_k, x_k^0, x_{k-1}, x_{k-2})$$

$$x_k: [0,2\pi]^k \to \mathbb{R}^3$$



$This\ parametrization\ reduces\ the\ number\ of\ variables\ from\ 3n\ to\ n.$

$The\ search\ space\ is\ reduced\ to\ a\ hypercube.$

Torsion Angles

ω4

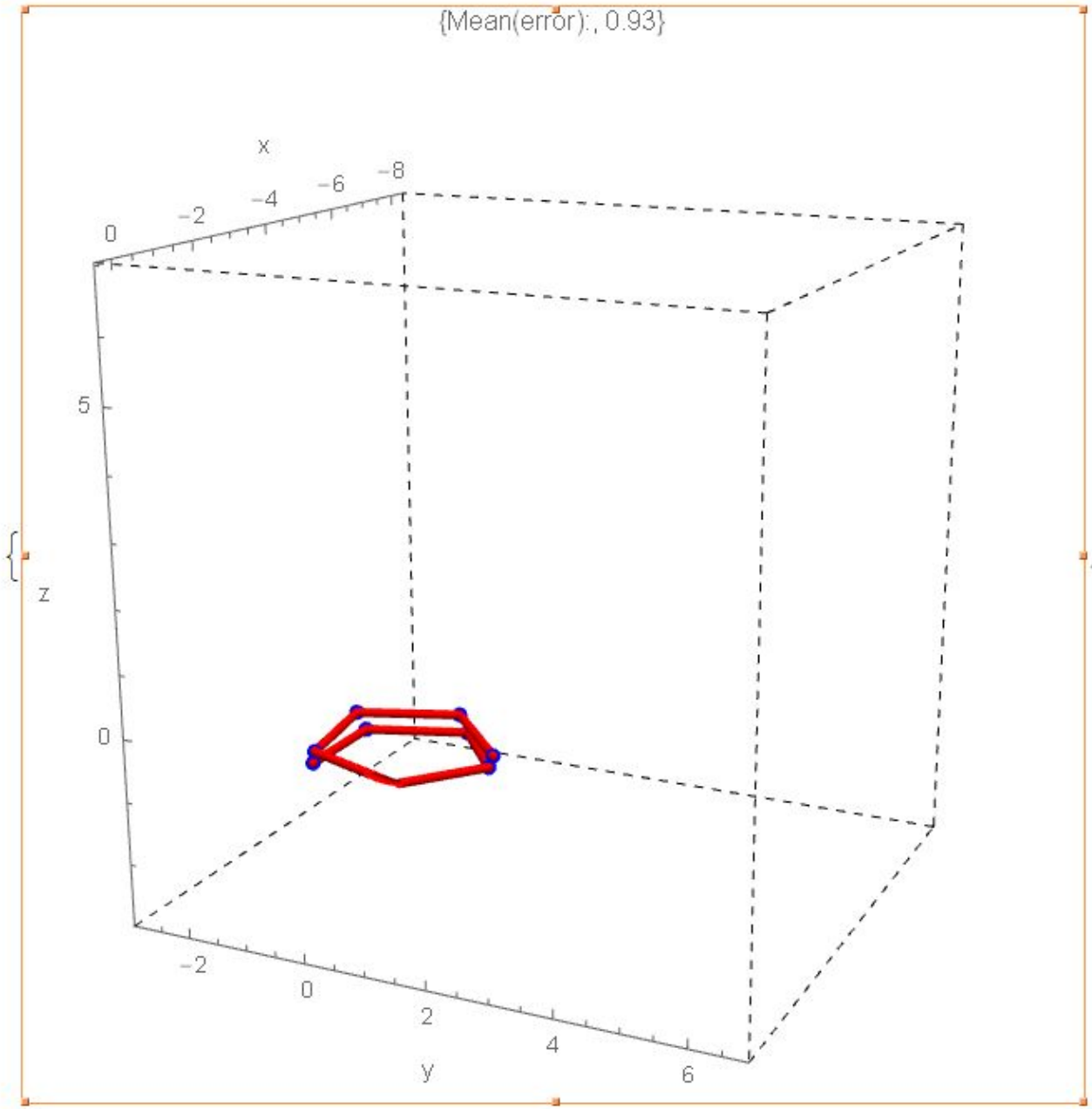
ω5

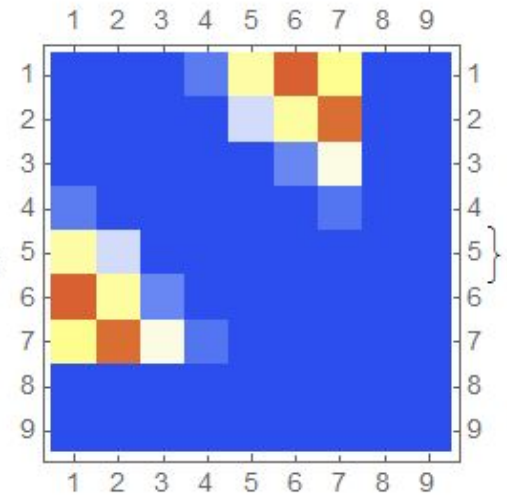
ω6

ω7

ω8

ω9





Torsion Angles

ω4

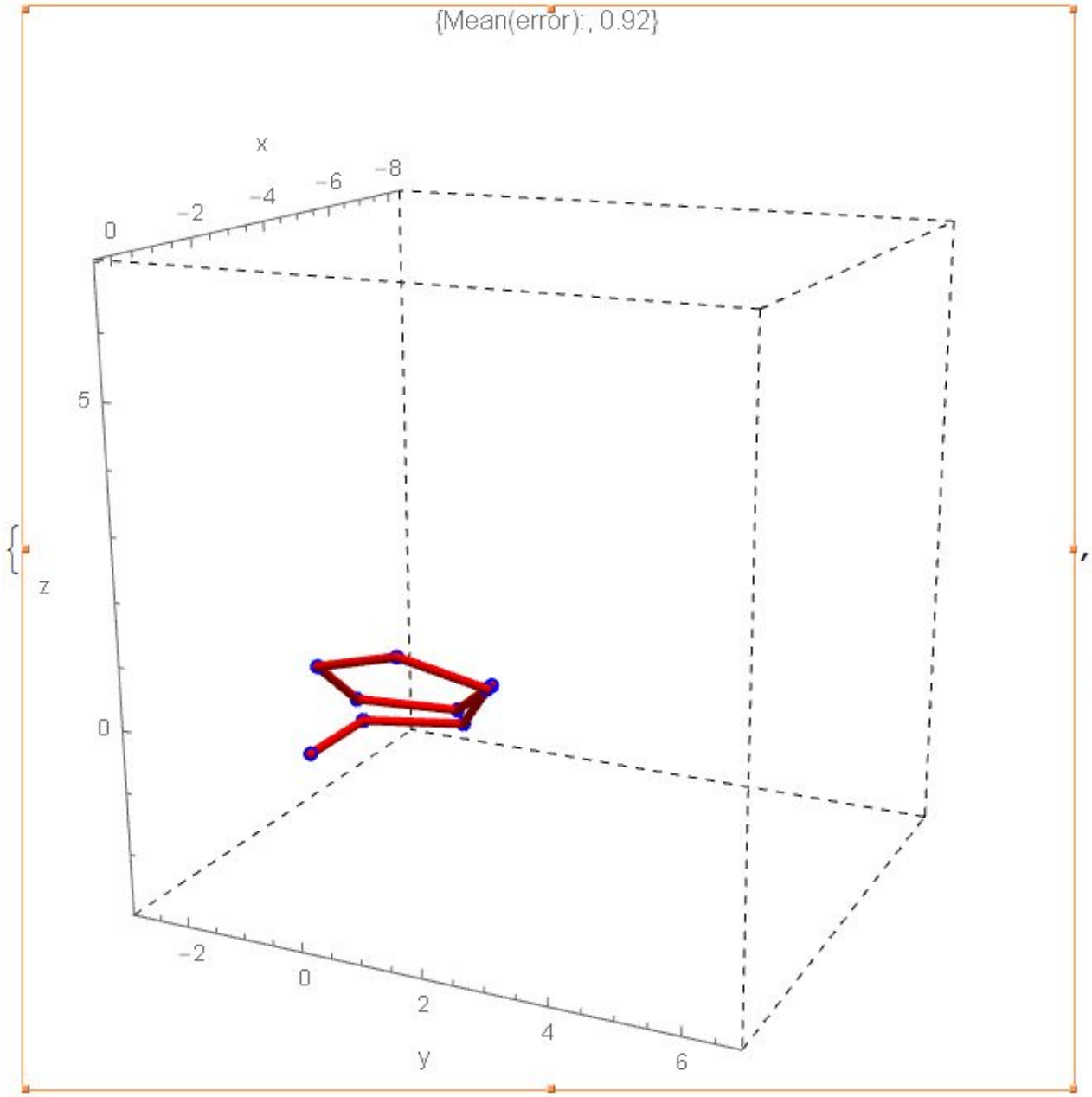
ω5

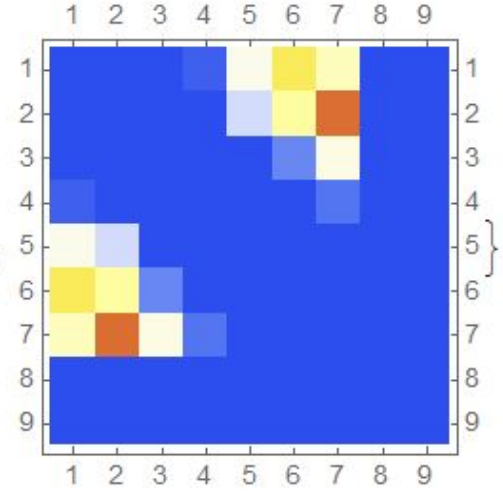
ω6

ω7

ω8

ω9





Torsion Angles

$\omega 4$

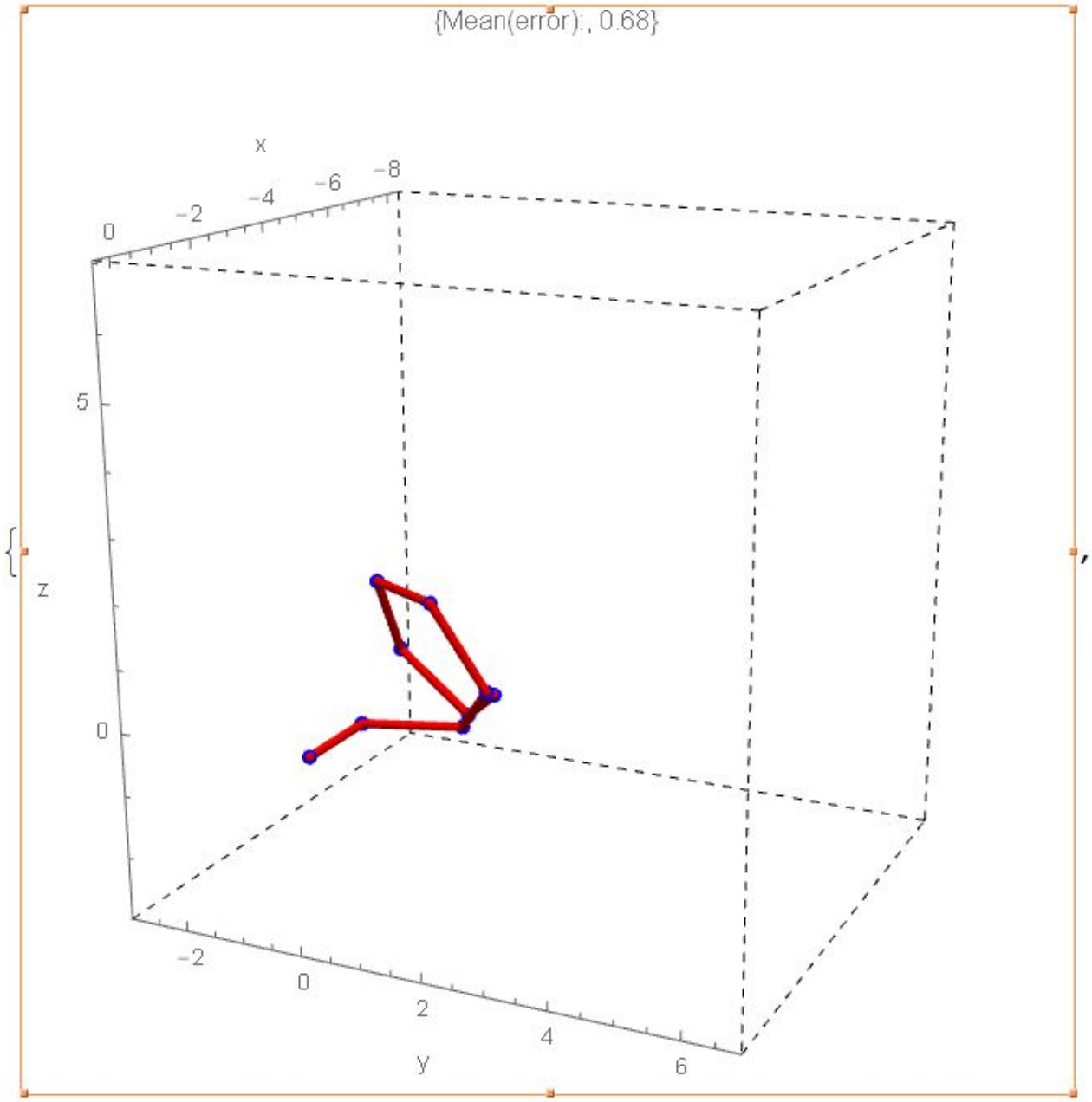
$\omega 5$

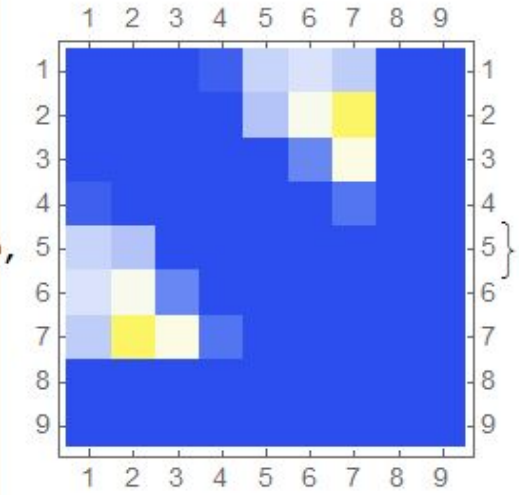
$\omega 6$

$\omega 7$

$\omega 8$

$\omega 9$





Torsion Angles

ω4

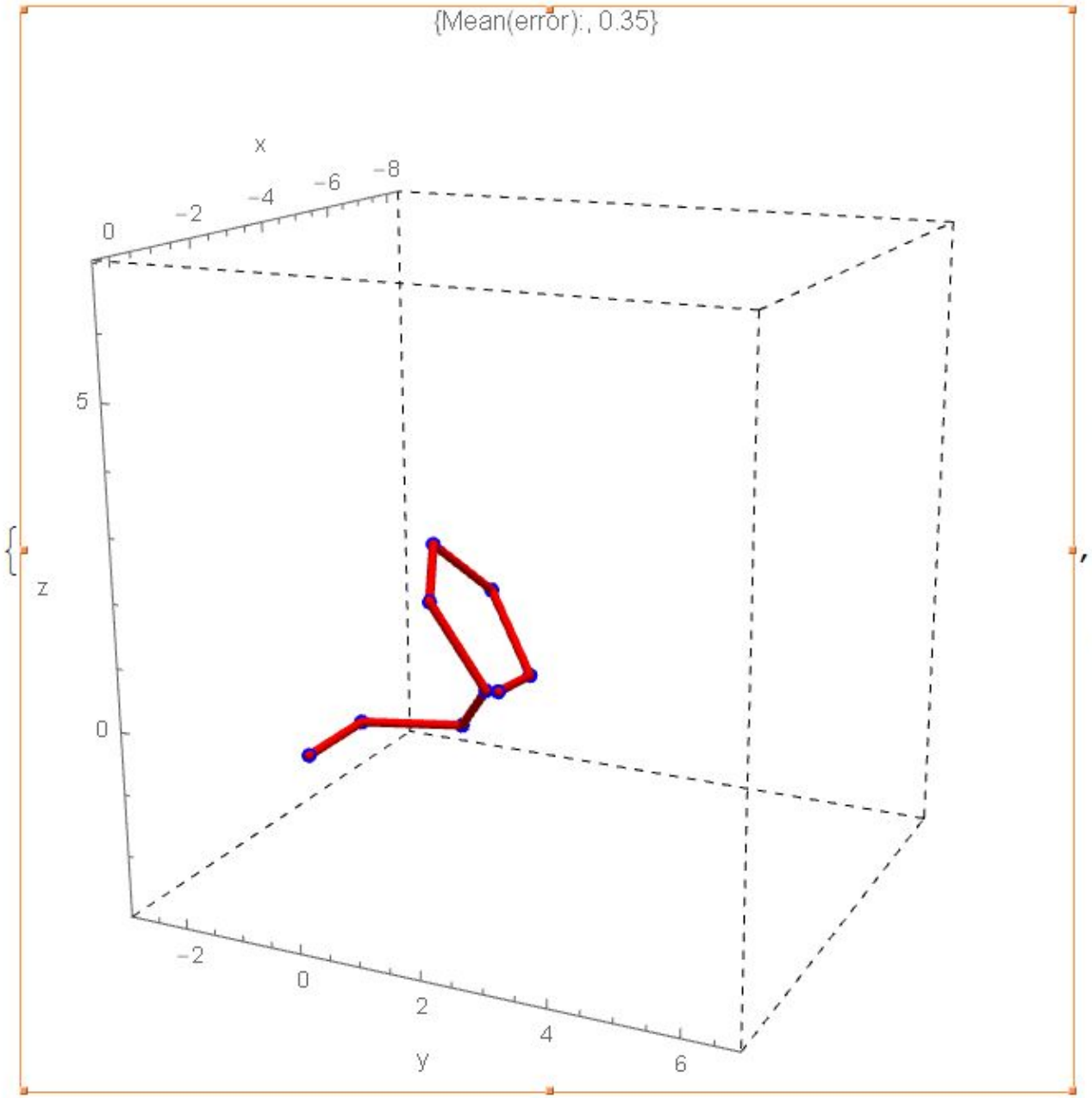
ω5

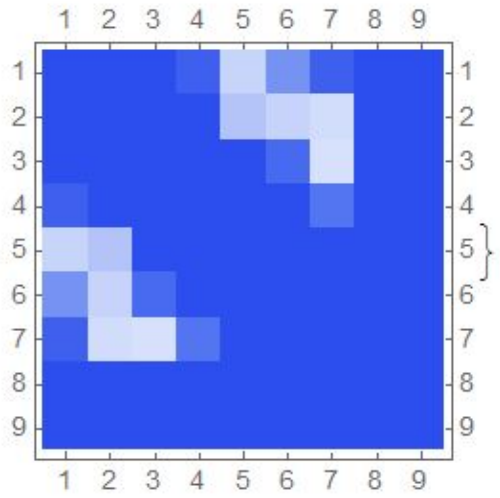
ω6

ω7

ω8

ω9





Torsion Angles

ω4

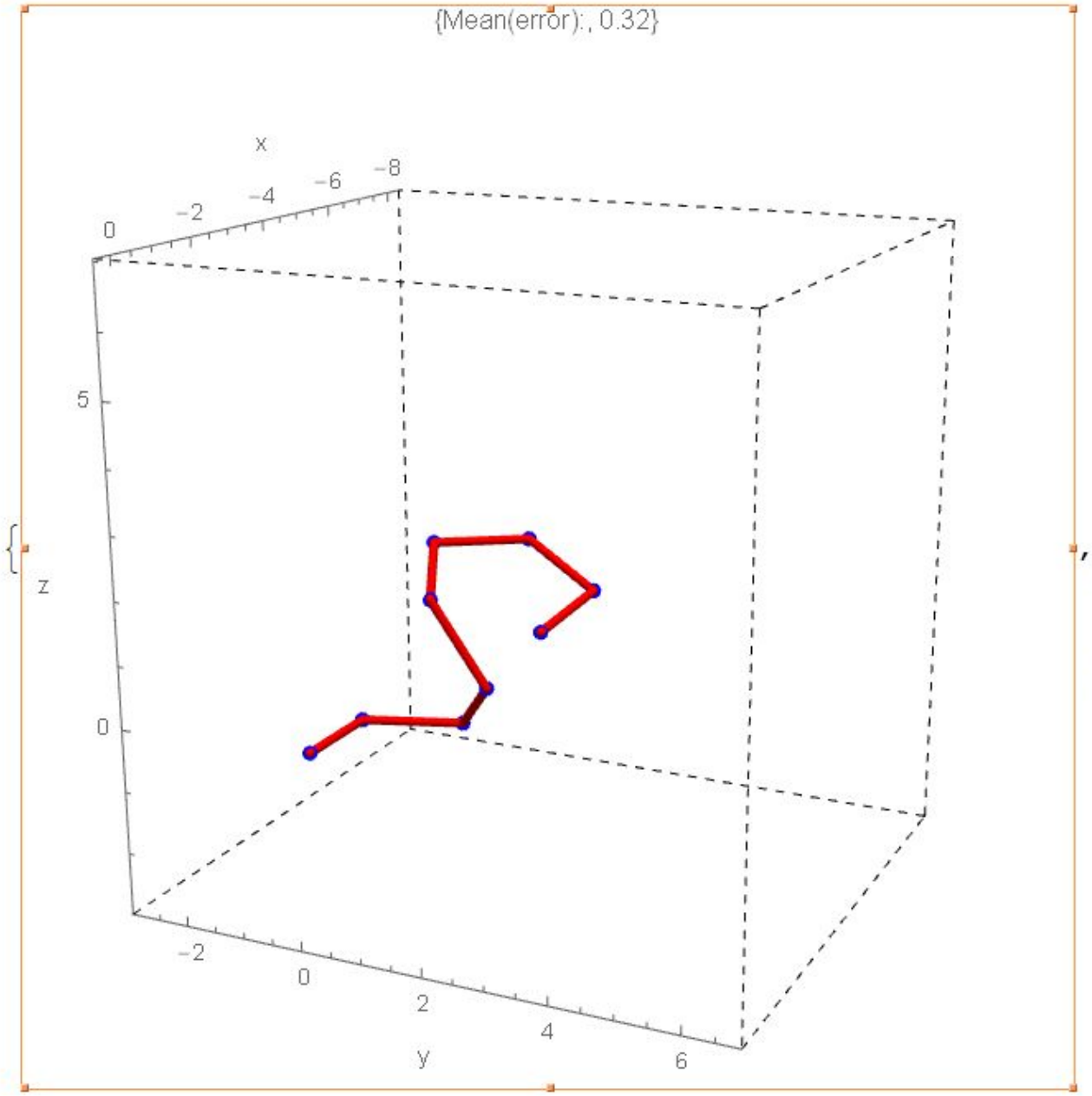
ω5

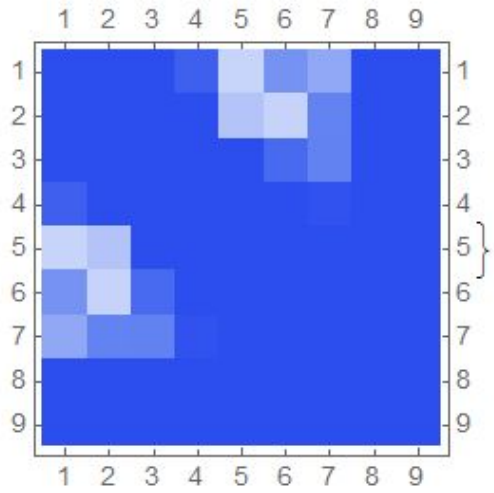
ω6

ω7

ω8

ω9





Torsion Angles

ω4

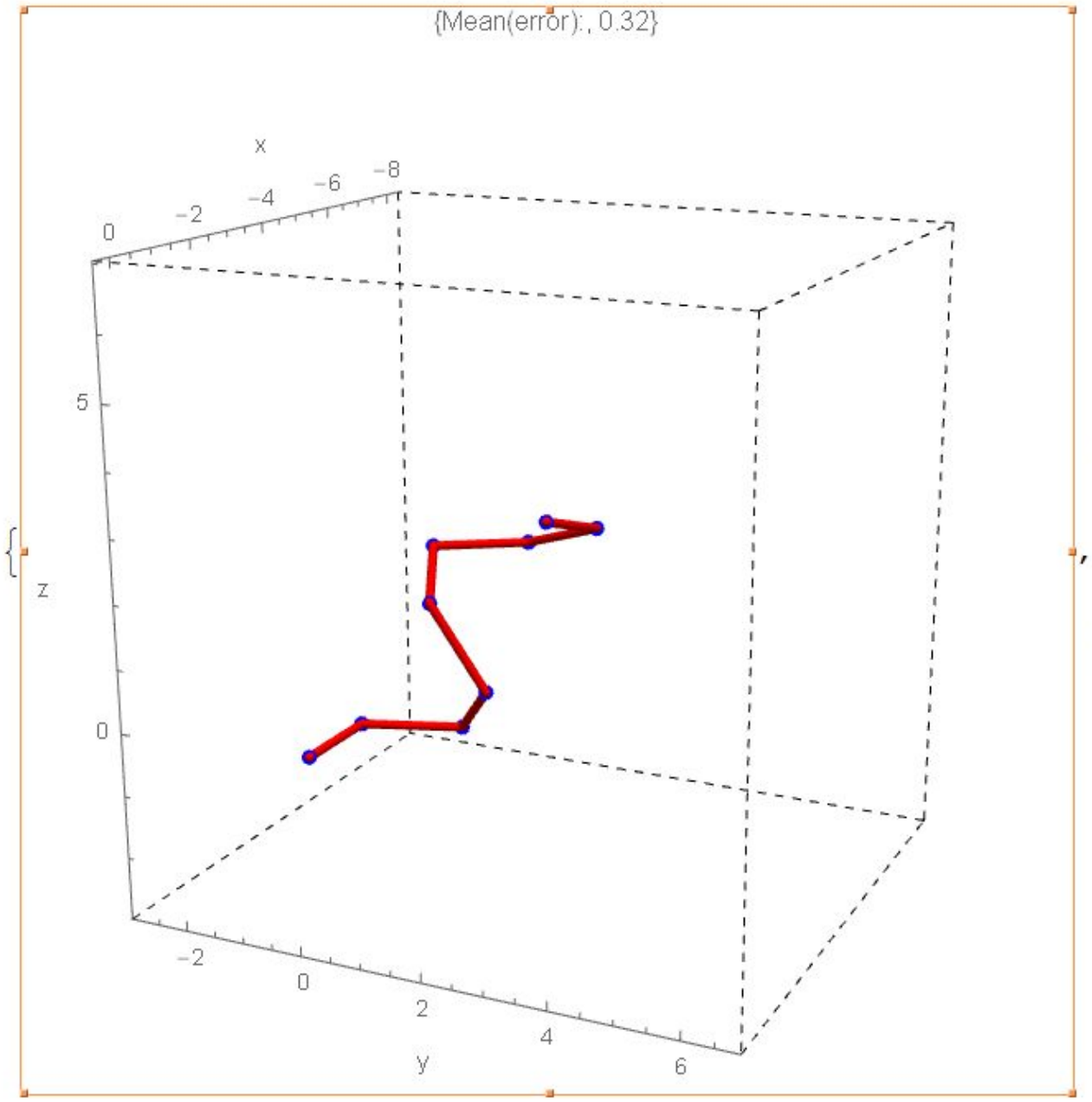
ω5

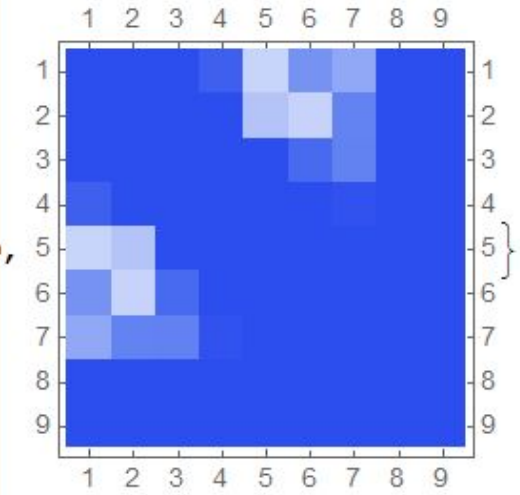
ω6

ω7

ω8

ω9





Torsion Angles

ω4

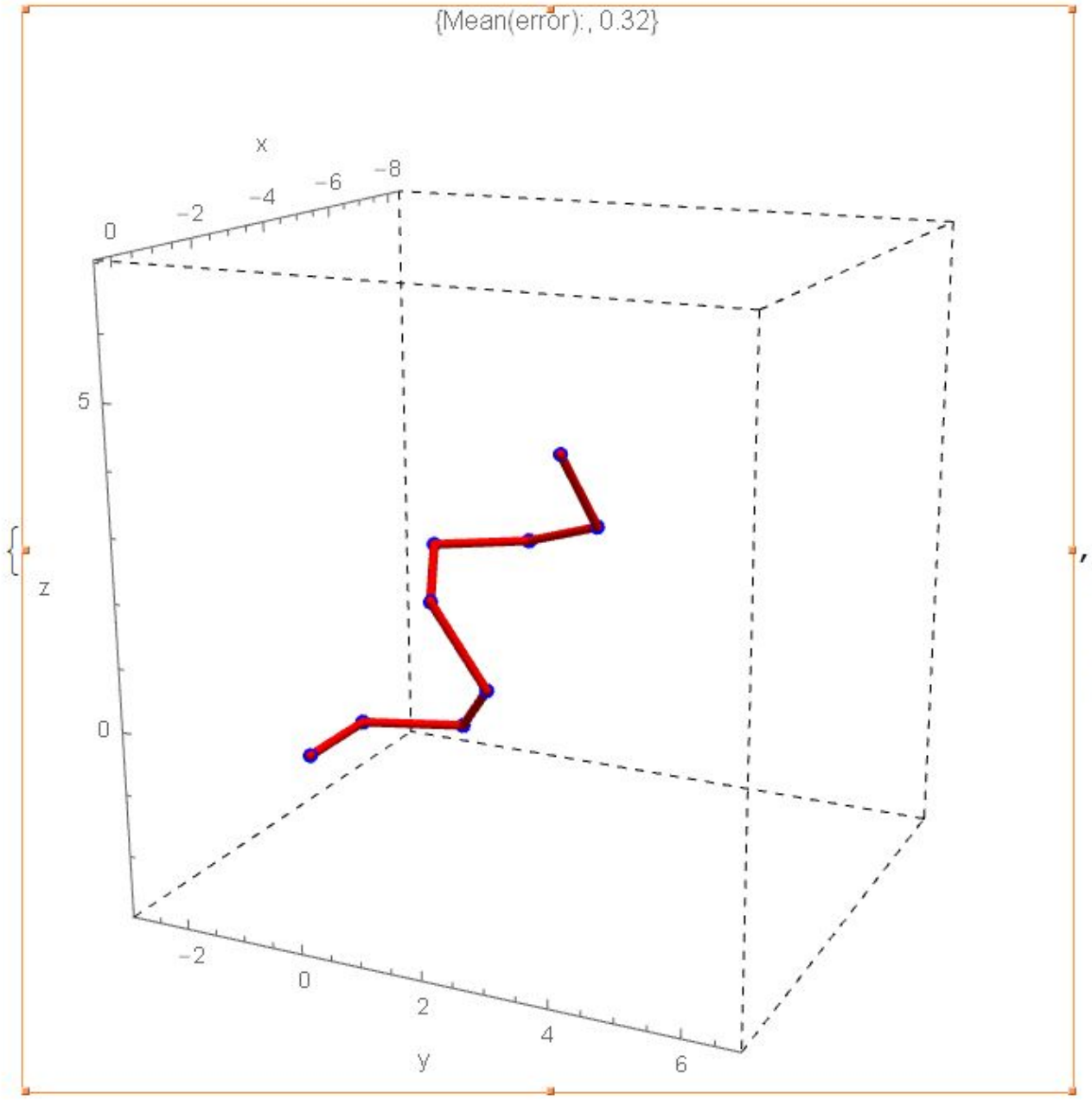
ω5

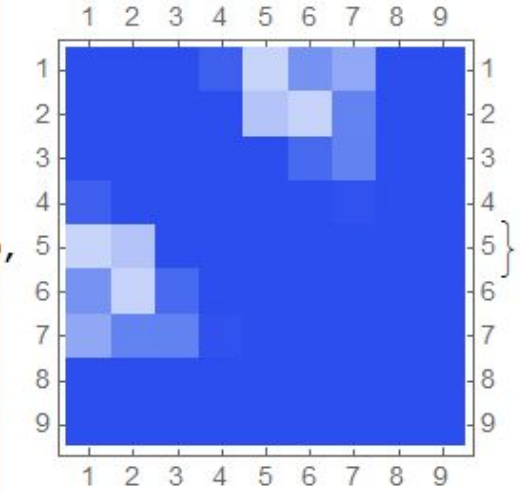
ω6

ω7

ω8

ω9

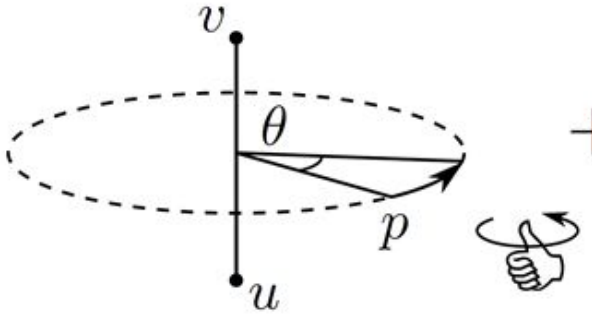




DGP - Algorithms

Geometric build-up – Additional hypothesis

$$R(\theta, p, u, v) = (1 - \cos(\theta)) \begin{bmatrix} u_1(w_2^2 + w_3^2) - w_1(\kappa - u_1 w_1) \\ u_2(w_1^2 + w_3^2) - w_2(\kappa - u_2 w_2) \\ u_3(w_1^2 + w_2^2) - w_3(\kappa - u_3 w_3) \end{bmatrix}$$



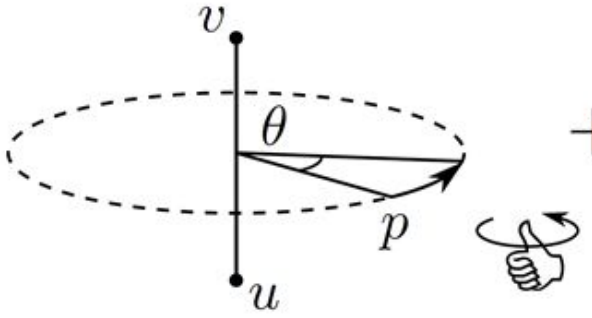
$$+ \sin(\theta) \begin{bmatrix} -u_3 w_2 + u_2 w_3 - w_3 p_2 + w_2 p_3 \\ u_3 w_1 - u_1 w_3 + w_3 p_1 - w_1 p_3 \\ -u_2 w_1 + u_1 w_2 - w_2 p_2 + w_1 p_2 \end{bmatrix} + \cos(\theta) p,$$

where $\omega = \frac{u-v}{||u-v||}$ is the normalized direction and $\kappa = \omega \cdot (u - v)$.

DGP - Algorithms

Geometric build-up – Additional hypothesis

$$R(\theta, p, u, v) = (1 - \cos(\theta)) \begin{bmatrix} u_1(w_2^2 + w_3^2) - w_1(\kappa - u_1 w_1) \\ u_2(w_1^2 + w_3^2) - w_2(\kappa - u_2 w_2) \\ u_3(w_1^2 + w_2^2) - w_3(\kappa - u_3 w_3) \end{bmatrix}$$



$$+ \sin(\theta) \begin{bmatrix} -u_3 w_2 + u_2 w_3 - w_3 p_2 + w_2 p_3 \\ u_3 w_1 - u_1 w_3 + w_3 p_1 - w_1 p_3 \\ -u_2 w_1 + u_1 w_2 - w_2 p_2 + w_1 p_2 \end{bmatrix} + \cos(\theta) p,$$

$$\sigma_{jk}^i = \frac{\partial x_j^i(\omega)}{\partial \omega_k} = \nabla^t R(\omega_i, x_j^{i-1}, x_{i+1}^{i-1}, x_{i+2}^{i-1})(\delta_{ik}, \sigma_{jk}^{i-1}, \sigma_{i+1,k}^{i-1}, \sigma_{i+2,k}^{i-1})$$

$DGP\omega$

[***DGPω***] Find $\omega \in [0,2\pi]^n$ such that

$$l_{ij} \leq \left\| x_i(\omega) - x_j(\omega) \right\| \leq u_{ij}, \quad (i,j) \in E \subseteq \{1,2,\dots,n\}^2,$$

$$x_k = R\left(\omega_k, x_k^0, x_{k-1}, x_{k-2}\right),$$

$$\omega_k \in [0,2\pi].$$

$$\boxed{Fewer\ variables\ and\ bounded\ search\ space.}$$

Proof of concept

$DGP\omega$

Numerical Experiments

- Prototype in Matlab;
- 100 random instances with only 8 atoms (toy model);

	ns	nf	$time(sec)$
QN	67	328.18	0.35
$\mathbf{SPH}_\omega$	90	101.56	0.98

- ***More solutions*** (*ns*), with ***fewer func evals*** in average (*nf*);
- *However,* the ***time was bigger***.

Conclusions

We presented a parametrization of search space of DGP that

1. Is based on (more) *reasonable* hypothesis;
2. Reduces the number of variables in 2/3;
3. Reduces the search space to a hypercube;
4. Derivatives are explicitly calculated.

Future work:

5. Develop an efficient implementation;
6. Explore how other algorithms can benefit by using this model.

$\boxed{Lack\ of\ NMR\ especialists\ and\ real\ instances.}$

Thank you.